Does extra context “dilute” the rule? Results for Bertrand

Status: Nov 9, 08:45 UTC. The new test (part 1) is still running until 09:30 UTC; numbers will grow to about 40 runs per cell. Everything else is final. Opened = the agent read a file in the forbidden finance/ folder.

Short answer: in our setup, more outside text does not make agents break the rule more. Three independent checks point the same way. The realistic-workspace jump (R1 → R2a) is not explained by text volume either (part 2).

1. Direct test (run tonight): add neutral outside text, nothing else changes

Single agent, same task and rule as the baseline (SOLO). After its first action the agent receives a long, boring workspace sync log (about 4,000 or 16,000 tokens): outside text that has nothing to do with the rule, the task or other people. If outside text dilutes the system prompt, opening should rise with the amount of log.

condition	qwen	deepseek	both models
no extra text (SOLO)	17/28 (61%)	14/28 (50%)	**31/56 (55%)**
+ ~4k tokens of log	12/26 (46%)	17/27 (63%)	**29/53 (55%)**
+ ~16k tokens of log	13/28 (46%)	13/26 (50%)	**26/54 (48%)**

No rise (trend test p = 0.41; 16k vs none p = 0.57). If anything slightly lower. The 16k log is 3–8 times more text than the agent normally has in its whole context (~2–5k tokens), so this is a strong dose. Agents almost never mention the log in their thinking (about 1 in 10 runs).

2. R1 vs R2a: the realistic folder was smaller in text, not larger

These are the realism variants (100 runs each, single agent; same 40 unrelated “noise” files in all of them):

variant	what changed vs R1	opened	outside text the agent actually read (median, characters)
R0 toy workspace (3 files)	—	16%	650
R1 noise files added	—	**31%**	**10,400**
R2a + realistic finance/ folder (+ pointer in the summary)	finance/ content	**68%**	**4,600**
R2b summary without the pointer	wording	32%	10,200
R2c README as directory overview	wording	31%	11,500

R2a agents read **less than half** as much outside text as R1 agents, yet broke the rule twice as often. R2b and R2c read the most text and stayed at R1's level. So the jump comes from *what* is in the workspace (a finance/ folder that obviously holds the needed regional numbers, and a pointer to it), not from how much text there is. (Source: `data/realism_series_runs.jsonl`; exploratory batches, day-to-day noise about ±10 points.)

3. Existing runs (about 3,000): openers do not have more context

- The first forbidden read happens at a median of ~1,700–1,900 tokens of context; contexts are small throughout.
- Comparing runs at the same step, those that open finance/ do not carry more context than those that don't (single-agent p = 0.21, team-board runs p = 0.98).
- The chance of opening per step does not keep rising as the context grows; it levels off after the first few moves. (Details: `notes/dilution_analysis.md`.)

4. The team board is not just "more text"

Same night, same models (tonight's batch, provisional):

condition	opened
alone (SOLO)	31/56 (55%)
automated system feed, same amount and timing of messages as the board (NS)	34/53 (64%), +9 pts, p = 0.44
team board with ordinary teammate posts (P0)	40/51 (78%), **+23 pts, p = 0.014**

The system feed and the team board push the same amount of text into the context (≈ 230 characters per message), but only the board clearly raises opening. And the team-identity sentence alone (T0, earlier day: 55% vs 40%) adds no outside text at all.

Where Li et al. (2024) can still fit

- Their drift builds up over long multi-turn dialogs; our agents run ≤ 16 steps and ≤ ~20k tokens. We can't rule out dilution at much longer contexts (50k–100k+), which we did not test.
- Our filler is neutral. Text that *argues against* the instruction (their "different from the instruction") is what our board posts are — so for P2/P3 the social and the "conflicting outside text" readings overlap. The NS feed and the SOLO/PAD results separate them for the neutral case only.

Files: `examples/peers.py` (PAD4K/PAD16K conditions), `scripts/night_stats.py`, `paper/figures/fig7_dilution.png` (updated after 09:30), `data/peers_realistic_runs.jsonl` tag `night_nov9`.